

Variational Inference Problems

ELG 5218 – Uncertainty Evaluation in Engineering Measurements and Machine Learning
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PART A: CONCEPTUAL QUESTIONS (Simple to Intermediate)

A1. What does Shannon entropy measure, and why is the uniform distribution the maximum entropy distribution over a finite support?

A2. Why is the KL divergence not a true distance metric? State at least two properties it lacks.

A3. Explain the difference between forward KL $\text{KL}(p\|q)$ and reverse KL $\text{KL}(q\|p)$ in terms of their mode-covering vs. mode-seeking behavior. What happens when the true distribution p is bimodal and q is a single Gaussian?

A4. Why does variational inference minimize the reverse KL $\text{KL}(q\|p(z | x))$ rather than the forward KL $\text{KL}(p(z | x)\|q)$?

A5. What is the mean-field approximation? State its key assumption and describe its main limitation.

PART B: MATHEMATICAL DERIVATIONS

B1. Derive the Evidence Lower Bound (ELBO) using Jensen's inequality.

Problem. Given observed data x , latent variables z , joint distribution $p(x, z)$, and a variational distribution $q(z)$, start from

$$\log p(x) = \log \int p(x, z) dz$$

and derive the ELBO:

$$\mathcal{L}(q) = \mathbb{E}_{q(z)}[\log p(x, z)] - \mathbb{E}_{q(z)}[\log q(z)].$$

B2. Show that maximizing the ELBO is equivalent to minimizing $\text{KL}(q(z)\|p(z | x))$.

Problem. Starting from the definition of $\text{KL}(q(z)\|p(z | x))$, show that

$$\log p(x) = \mathcal{L}(q) + \text{KL}(q(z)\|p(z | x)),$$

and explain why this implies the equivalence.

B3. Derive the alternative form of the ELBO.

Problem. Show that the ELBO can be written as

$$\mathcal{L}(q) = \mathbb{E}_{q(z)}[\log p(x | z)] - \text{KL}(q(z) \| p(z)),$$

and interpret each term.

PART C: PARAMETRIC ANALYSIS (What if we change parameters?)

C1. Consider the alternative ELBO form $\mathcal{L}(q) = \mathbb{E}_{q(z)}[\log p(x | z)] - \text{KL}(q(z) \| p(z))$. What happens to the optimal variational distribution $q^*(z)$ as the number of observed data points N increases? Discuss the relative importance of the two terms.

C2. In mean-field variational inference with $q(z) = \prod_i q_i(z_i)$, what happens to the quality of the approximation as the true posterior $p(z | x)$ becomes more strongly correlated?

PART D: IMPLEMENTATION AND PRACTICAL QUESTIONS

D1. What is the reparameterization trick? Explain why naive Monte Carlo sampling breaks gradient computation and how the trick solves this.

D2. Compare the score-function (REINFORCE) estimator and the pathwise (reparameterization) estimator for computing $\nabla_{\lambda} \mathcal{L}(\lambda)$. Under what conditions would you prefer one over the other?

D3. In the coordinate ascent algorithm for mean-field VI, why must the updates be performed cyclically rather than simultaneously? What guarantees convergence?

PART E: COMPUTATION PROBLEMS

E1: KL divergence between two Bernoulli distributions.

Problem. Let $P = \text{Ber}(0.8)$ and $Q = \text{Ber}(0.5)$.

- (a) Compute $\text{KL}(P \| Q)$ in nats.
- (b) Compute $\text{KL}(Q \| P)$ in nats.
- (c) Verify that $\text{KL}(P \| Q) \neq \text{KL}(Q \| P)$.

E2: Entropy and ELBO for a 1D Gaussian model.

Problem. Consider the model $p(z) = \mathcal{N}(0, 1)$, $p(x | z) = \mathcal{N}(x; z, 1)$, with observed $x = 3$. Let the variational distribution be $q(z) = \mathcal{N}(z; \mu, \sigma^2)$.

- (a) Write the ELBO $\mathcal{L}(\mu, \sigma)$ as a closed-form function of μ and σ .
- (b) Find the optimal μ^* and σ^{*2} that maximize the ELBO.
- (c) What is the true posterior $p(z | x)$? Compare with your answer in (b).

E3: Entropy computations.

Problem.

- (a) Compute the Shannon entropy (in bits) of a discrete distribution $P = (0.5, 0.25, 0.125, 0.125)$.
- (b) Show that the differential entropy of $X \sim \mathcal{N}(\mu, \sigma^2)$ is $h(X) = \frac{1}{2} \log(2\pi e \sigma^2)$.